# Gilberto Díaz-García

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## Research Interests

Design & Analysis of Large-Scale Systems, Distributed Control, Game Theory and its Control Engineering Applications.

## Education

**Ph.D. in Electrical & Computer Engineering** — *Santa Barbara, California, USA*
UNIVERSITY OF CALIFORNIA, SANTA BARBARA — *2021 (Ongoing)*

**M.Sc. in Electronic & Computer Engineering** — *Bogotá, Colombia*
UNIVERSITY OF LOS ANDES - *Cum Laude*, GPA: 4.86/5.0 — *2018–2020*
Thesis: "Resilient Synchronization of Multi-agent Systems: A Game Theoretical Approach".
Advisor: Luis Felipe Giraldo.

**B.Sc. in Electronic Engineering** — *Bogotá, Colombia*
UNIVERSITY OF LOS ANDES - GPA: 4.4/5.0 — *2014–2018*

**Minor in Computational Mathematics** — *Bogotá, Colombia*
UNIVERSITY OF LOS ANDES — *2014–2018*
A structured interdisciplinary option with undergraduate courses in Computers and Systems Engineering and Mathematics.

## Journal Articles

- **G. Díaz-García**, F. Bullo, and J. R. Marden, "Strategic Coalitions in Networked Contest Games," *IEEE Transactions on Automatic Control*, 2025
- Y. John, **G. Díaz-García**, X. Duan, J. R. Marden, and F. Bullo, "A Stochastic Surveillance Stackelberg Game: Co-Optimizing Defense Placement and Patrol Strategy," *IEEE Transactions on Automatic Control*, 2025
- **G. Díaz-García**, F. Bullo, and J. R. Marden, "Distributed Markov Chain-Based Strategies for Multi-Agent Robotic Surveillance," *IEEE Control Systems Letters*, vol. 7, pp. 2527–2532, 2023
- J. Martinez-Piazuelo, **G. Diaz-Garcia**, N. Quijano, and L. F. Giraldo, "Discrete-time Distributed Population Dynamics for Optimization and Control," 11, vol. 52, IEEE, 2022, pp. 7112–7122
- **G. Díaz-García**, G. Narváez, L. F. Giraldo, J. Giraldo, and A. A. Cardenas, "Resilient Structural Sparsity in the Design of Consensus Networks," *IEEE Transactions on Cybernetics*, 2021
- **G. Díaz-García**, D. Guatibonza, and L. F. Giraldo, "Distributed Reconfiguration for Resilient Synchronization of Multi-Agent Systems," *IEEE Access*, vol. 9, pp. 140 235–140 247, 2021
- A. F. Zambrano, **G. Díaz-García**, S. Ramírez, L. F. Giraldo, H. Gonzalez-Villasanti, M. T. Perdomo, I. D. Hernández, and J. M. Godoy, "Donation Networks in Underprivileged Communities," *IEEE Transactions on Computational Social Systems*, pp. 1–10, 2020

## Conference Publications

- **G. Díaz-García**, K. Paarporn, and J. R. Marden, "When More Information Means Less: A Case Study in Asymmetric All-Pay Auctions," in *2025 IEEE 64th Conference on Decision and Control (CDC)*, IEEE, 2025, pp. 5387–5392
- **G. Díaz-García**, K. Paarporn, and J. R. Marden, "The Value of Compromising Strategic Intent in General Lotto Games," in *2025 American Control Conference (ACC)*, IEEE, 2025, pp. 1554–1559
- Y. John, C. Hughes, **G. Diaz-Garcia**, J. R. Marden, and F. Bullo, "RoSSO: A High-Performance Python Package for Robotic Surveillance Strategy Optimization Using JAX," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2024, pp. 2169–2175
- **G. Díaz-García**, F. Bullo, and J. R. Marden, "Beyond the 'Enemy-of-my-Enemy' Alliances: Coalitions in Networked Contest Games," in *2023 62nd IEEE Conference on Decision and Control (CDC)*, IEEE, 2023, pp. 2220–2225
- **G. Díaz-García**, C. A. Uribe, and N. Quijano, "Population Dynamics for Discrete Wasserstein Gradient Flows over Networks," in *International Conference on Machine Learning Conference: LatinX in AI (LXAI) Research Workshop. [Oral Presentation]*, 2021, Virtual
- **G. Díaz-García**, L. F. Giraldo, and S. Jimenez-Leudo, "Dynamics of a Differential Wheeled Robot: Control and Trajectory Error Bound," in *2021 IEEE 5th Colombian Conference on Automatic Control (CCAC)*, IEEE, 2021, pp. 25–30
- J. Martinez-Piazuelo, **G. Diaz-Garcia**, N. Quijano, and L. F. Giraldo, "Distributed Formation Control of Mobile Robots using Discrete-Time Distributed Population Dynamics," in *Proceedings of the 21st IFAC World Congress*, vol. 53, Elsevier, 2020, pp. 3131–3136
- **G. Díaz-García**, L. Burbano, N. Quijano, and L. F. Giraldo, "Distributed MPC and Potential Game Controller for Consensus in Multiple Differential-Drive Robots," in *Proceedings of the 2019 IEEE 4th Colombian Conference on Automatic Control (CCAC)*, IEEE, 2019, pp. 1–6

## Teaching Experience

**Instructor & Lead Researcher** — *Santa Barbara, California, USA*
Summer Research Academies. Track #1: Complex Systems — *Jun 2025 - Jul 2025*
University of California, Santa Barbara

**Graduate Teaching Assistant** — *Santa Barbara, California, USA*
Game Theory for Networked Systems — *Jan 2025 - Mar 2025*
University of California, Santa Barbara

**Graduate Teaching Assistant** — *Santa Barbara, California, USA*
Digital Control Systems - Theory and Design — *Jan 2024 - Mar 2024*
University of California, Santa Barbara

**Graduate Teaching Assistant** — *Santa Barbara, California, USA*
Feedback Control Systems - Theory and Design — *Sept 2022 - Dec 2022*
University of California, Santa Barbara

**Graduate Teaching Assistant** — *Santa Barbara, California, USA*
Signal Analysis and Processing — *Jan 2022 - Mar 2022*
University of California, Santa Barbara

**Graduate Teaching Assistant** — *Bogotá, Colombia*
Control Systems Analysis — *Aug 2018 – Nov 2018*
University of Los Andes

**Graduate Teaching Assistant** — *Bogotá, Colombia*
Stochastic Processes — *Jan 2018 – Jun 2018*
University of Los Andes

**Graduate Teaching Assistant** — *Bogotá, Colombia*
OPTIMIZATION — *Jan 2018 – Jun 2018*
University of Los Andes

**Undergraduate Teaching Assistant** — *Bogotá, Colombia*
COURSE ON SCIENTIFIC COMPUTING FOR ENGINEERING — *Aug 2017 – Nov 2017*
University of Los Andes

**Undergraduate Teaching Assistant** — *Bogotá, Colombia*
CONTROL SYSTEMS ANALYSIS — *Aug 2016 – Nov 2017*
University of Los Andes

**Undergraduate Teaching Assistant** — *Bogotá, Colombia*
ALGORITHMIC AND OBJECT ORIENTED PROGRAMMING II (HONORS) — *Aug 2016 – Nov 2016*
University of Los Andes

**Undergraduate Teaching Assistant** — *Bogotá, Colombia*
ALGORITHMIC AND OBJECT ORIENTED PROGRAMMING I — *Jan 2015 – Jun 2016, Jan 2017 - Jun 2017*
University of Los Andes

## Research Experience

**Graduate Research Assistant** — *Santa Barbara, California, USA*
CENTER FOR CONTROL, DYNAMICAL SYSTEMS, AND COMPUTATION — *Jan 2022 – Dec 2025*
University of California, Santa Barbara.

**Graduate Research Assistant** — *Bogotá, Colombia*
COMPUTATIONAL MODELS OF COOPERATION IN UNDERPRIVILEGED COMMUNITIES — *Jan 2019 – Nov 2020*
University of Los Andes. Funded through Google Latin America Research Awards.

## Reviewing Experience

### International Journals

- IEEE Control Systems Letters.
- IEEE Transactions on Neural Networks and Learning Systems.
- IEEE Transactions on Systems, Man and Cybernetics: Systems.
- IEEE Transactions on Cybernetics.
- IEEE Control Systems Letters.
- International Journal of Robust and Nonlinear Control.
- Peer-to-Peer Networking and Applications.
- Mathematics of Control, Signals, and Systems.
- Applied Network Science.
- Nonlinear Dynamics.
- Social Network Analysis and Mining.

### International Conferences

- European Control Conference, 2025, 2024.
- American Control Conference, 2025, 2024.
- IEEE Conference on Decision and Control, 2025, 2023.
- IFAC World Congress, 2023, 2020.
- IEEE Conference on Control Technology and Applications, 2020.
- IEEE Colombian Conference on Automatic Control, 2019.

## Academic Awards & Honors

**Neal Fenzi - Resonant Founder Fellowship** *Santa Barbara, California, USA*

UNIVERSITY OF CALIFORNIA, SANTA BARBARA - DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING *Apr 2025*

Award in recognition of outstanding academic record and scholarship potential. Fellowship in honor of the retirement of Resonant, Inc. co-founder, Neal Fenzi.

**M.Sc. in Electronic and Computer Engineering Cum Laude Honor** *Bogotá, Colombia*

UNIVERSITY OF LOS ANDES *Mar 2020*

Honor for high academic performance. Top 3% GPA in M.Sc. in Electronic and Computer Engineering class for the past five years.

**Google Latin America Research Awards** *Bello Horizonte, Brazil*

GOOGLE LATIN AMERICA *Nov 2018 & Nov 2019*

Project: "Computational Models of Cooperation in Underprivileged Communities".

**"Bachilleres Por Colombia - Mario Galán Gómez" Award** *Bogotá, Colombia*

ECOPETROL S.A. *Sept 2013*

Scholarship for undergraduate studies in Colombia. Award for the best high-school students in Colombia.

**"Los Mejores en Educación" Award** *Bogotá, Colombia*

MINISTRY OF EDUCATION *Dic 2012*

Award for the high-school students with highest grades in ICFES national test in Colombia.

## Additional Skills

**Languages** English (Fluent), Spanish (Native)

**Programming Languages** Matlab & Simulink, Python, Java, C++